Example 1.31

Test for consistency of the following system of linear equations and if possible solve: $x-y+z=-9,\ 2x-2y+2z=-18,\ 3x-3y+3z+27=0.$

Solution

Here the number of unknowns is 3.

The matrix form of the system is $AX=B,$ where

$$A=\begin{bmatrix}1&-1&1\\2&-2&2\\3&-3&3\end{bmatrix},\ X=\begin{bmatrix}x\\y\\z\end{bmatrix},\ B=\begin{bmatrix}-9\\-18\\-27\end{bmatrix}.$$

Applying elementary row operations on the augmented matrix $[A\,|\,B]$, we get

$$[A\,|\,B]=\left[\begin{array}{ccc|c}1&-1&1&-9\\2&-2&2&-18\\3&-3&3&-27\end{array}\right]\xrightarrow{\substack{R_2\to R_2-2R_1,\\ R_3\to R_3-3R_1}}\left[\begin{array}{ccc|c}1&-1&1&-9\\0&0&0&0\\0&0&0&0\end{array}\right].$$

So, $\rho(A)=\rho([A\,|\,B])=1<3.$

From the echelon form, we get the equivalent equations $x-y+z=-9,\ \ 0=0,\ 0=0.$

The equivalent system has one non-trivial equation and three unknowns.

Taking $y=s, z=t$ arbitrarily, we get $x-s+t=-9;$ or $x=-9+s-t.$

So, the solution is $\left(x=-9+s-t, y=s, z=t\right),$ where s and t are parameters.

The above solution set is a two-parameter family of solutions.

Here, the given system of equations is consistent and has infinitely many solutions which form a two parameter family of solutions.

Example 1.32

Test the consistency of the following system of linear equations

$x-y+z=-9,\ 2x-y+z=4,\ 3x-y+z=6, 4x-y+2z=7.$

Solution

Here the number of unknowns is 3.

The matrix form of the system of equations is $AX=B,$ where

$$A=\begin{bmatrix}1&-1&1\\2&-1&1\\3&-1&1\\4&-1&2\end{bmatrix},\ X=\begin{bmatrix}x\\y\\z\end{bmatrix},\ B=\begin{bmatrix}-9\\4\\6\\7\end{bmatrix}.$$

Applying elementary row operations on the augmented matrix $[A\,|\,B]$, we get

$$[A\,|\,B]=\left[\begin{array}{ccc|c}1&-1&1&-9\\2&-1&1&4\\3&-1&1&6\\4&-1&2&7\end{array}\right]\xrightarrow{\substack{R_2\to R_2-2R_1,\\ R_3\to R_3-3R_1,\\ R_4\to R_4-4R_1}}\left[\begin{array}{ccc|c}1&-1&1&-9\\0&1&-1&22\\0&2&-2&33\\0&3&-2&43\end{array}\right]$$

$$\xrightarrow{\substack{R_3\to R_3-2R_2,\\ R_4\to R_4-3R_2}}\left[\begin{array}{ccc|c}1&-1&1&-9\\0&1&-1&22\\0&0&0&-11\\0&0&1&-23\end{array}\right]\xrightarrow{R_3\leftrightarrow R_4}\left[\begin{array}{ccc|c}1&-1&1&-9\\0&1&-1&22\\0&0&1&-23\\0&0&0&-11\end{array}\right]$$

So, $\rho(A)=3$ and $\rho([A\,|\,B])=4.$ Hence $\rho(A)\neq\rho([A\,|\,B]).$

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